Agentic AI with LangChain and LangGraph

Presentation 1: What’s AI Agent?
Primary Instructor: Michael Z
RA: YH
Date: 2026.Jan-Feb

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A little bit about me:

I am a RA of Dr. Michael.

I contributed to LangChain several times (their docs are open source so that everyone could contribute. My contribution got merged).

langchain-ai / docs

Code | Issues 145 | Pull requests 63 | Actions | Security | Insights

Remove wrong decision type from inline comment of HITL docs #1775

Merged mdrxy merged 1 commit into langchain-ai:main from Yacklin:correct-wrong-decision-type-in-HITL-docs 3 weeks ago

Conversation 0 | Commits 1 | Checks 12 | Files changed 1 +2 −2

Yacklin commented 3 weeks ago • edited Contributor

Overview

This PR removes decision type "edit" from an inline comment located in HITL docs. In the scenario described in the code snippet, only "approve" and "reject" are allowed but the inline comment includes "edit", which contradicts interrupt config.

Type of change

Type: Update existing documentation

Reviewers: Inhsingh

Assignees: No one assigned

Labels: langchain, oss

Fix missing type hints for Python script of Requesting Injectable Parameters and correct typo in Functional API docs #2054

Edit | Code

Merged mdrxy merged 2 commits into langchain-ai:main from Yacklin:patch-1 3 days ago

Conversation 1 | Commits 2 | Checks 12 | Files changed 1 +5 −2

Yacklin commented last week Contributor

Reviewers: Inhsingh

Correct typos and wrong information found in SummarizationMiddleware docs according to its reference for Python #2033

Edit | Code

Merged mdrxy merged 3 commits into langchain-ai:main from Yacklin:Yacklin-patch-1 1 hour ago

Conversation 1 | Commits 3 | Checks 12 | Files changed 1 +11 −11

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Not sure how many of you have heard the term “AI Agent”.

But I am curious about what you thought at the first time you knew this term?

The one like right hand side?

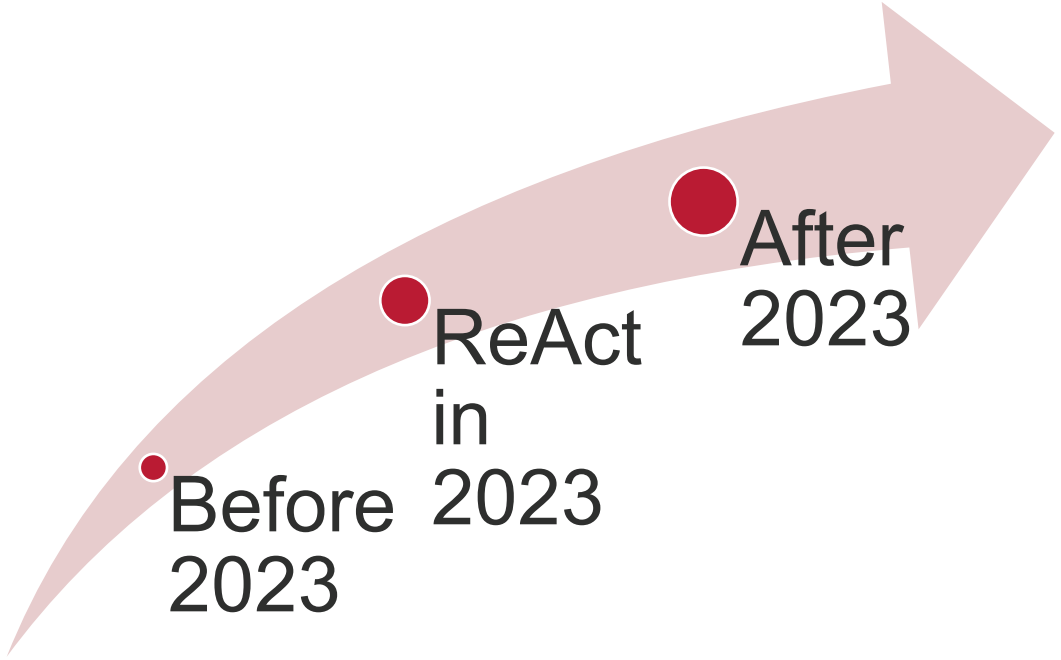
Or AI operated spy like the one below?

Figure 2: AI Spy

Figure 1: MarketWatch Illustration

Figure 1 Source: https://www.marketwatch.com/story/how-artificial-intelligence-could-change-the-way-you-buy-or-sell-your-next-house-579668be

Figure 2 Source: https://www.morphisec.com/blog/unpacking-the-cybersecurity-risks-of-ai-agent-based-systems/



0. A little bit of History

0. A little bit of History: Before 2023 (last century)

Decades ago, scientists started to imagine and define what agent should look like in the field of AI.

In 1995, Russell and Norvig said “agents operate autonomously, perceive their environments and persist over a prolonged time period, adapt to change and create and pursue goals” in the book *Artificial Intelligence: A Modern Approach* (first edition).

People in different fields, such as robotics, figured out what sort of agents they would like to research (not our focus though).

Scale things up: Craig Reynolds developed “boids” in 1986, showing that simple agent rules can produce global emergent behavior. This established the idea that societies of agents can produce intelligence exceeding the sum of their parts.

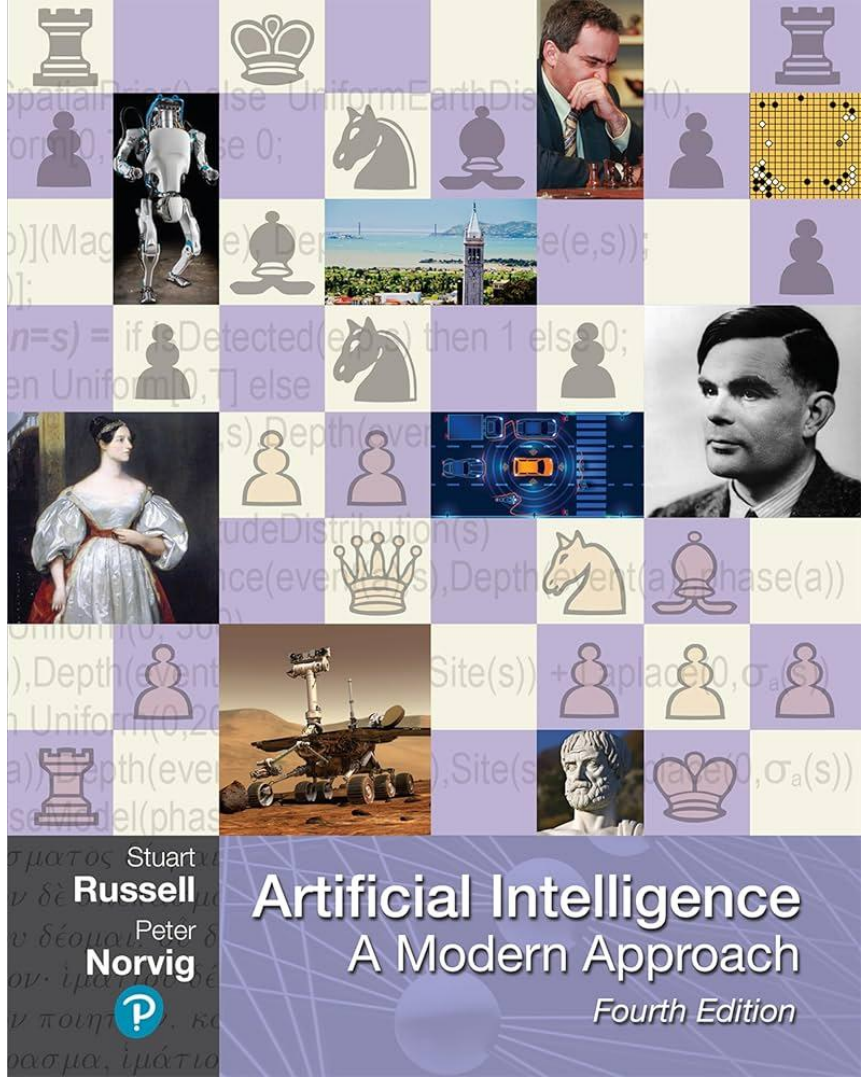


Figure 3: Cover of Book

0. A little bit of History: Before 2023 (Reinforcement Learning)

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Then, people developed AI system that could receive feedbacks and learn from the environments, to minimize penalty or maximize rewards (not our focus, even though it is also pretty interesting).

Notably, OpenAI and Google's AlphaGo (Do you want to watch these videos btw?):

Google's AI AlphaGo Is Beating Humanity At Its Own Games (HBO)
417K views • 8 years ago
VICE News
Go is an ancient, aristocratic Chinese board game that's reputed to have as many possible moves as there are atoms in the ...
CC
1 moments Which Google subsidiary developed the AlphaGo AI program that defeated a human professional Go...

Multi-Agent Hide and Seek
10M views • 6 years ago
OpenAI
We've observed **agents** discovering progressively more complex tool use while playing a simple game of **hide-and-seek**. Through ...
4K
6 moments Multiple Door Blocking | Ramp Use | Ramp Defense | Shelter Construction | Box Surfing | Surf Defense

0. A little bit of History: 2023

In 2022, Shunyu Yao wrote a paper in Google when working as an intern, which was accepted in ICLR 2023.

What makes this paper stand out from others is its potential: IT ACTUALLY CHANGES THE WORLD!!!

Two comments:

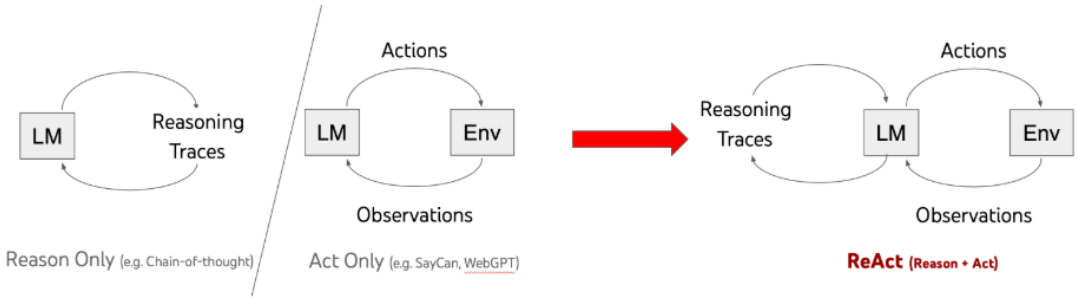
1. History of AI Agent development could be divided into two phases from now on, personally speaking: Pre-ReAct and ReAct.
2. All people are talking about agents after 2023 is basically based on ReAct Architecture.

(I am not sure if I should go over the paper for you, but this chart tells everything you need to know to be familiar with most of AI Agents building platform.)

ReAct: Synergizing Reasoning and Acting in Language Models

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, Yuan Cao

[Paper] [Code] [Blogpost] [BibTex]



About

I am a researcher at OpenAI.

I study agents.

Selected work

- **Computer-Using Agent (CUA)**
  OpenAI
  A universal agent/interface to interact with the digital world.
  product | blogpost | system card | bibtex

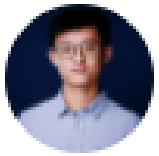
Quick Fun Facts:

1.There are two Shunyu Yao in USA and they both worked in AI industry. But the one who obtained PhD from Princeton is the creator of ReAct architecture (Many careless medias in one country confused and got confused for some reasons).
2.Shunyu Yao, I mean the one who created ReAct, became the Chief AI Scientist of Tencent, according to Bloomberg (Did you have the habit of reading Bloomberg News?).

Shunyu Yao • 3rd+
Senior Staff Research Scientist@Google DeepMind(ex@Anthropic)
Berkeley, California, United States

Past: Postdoctoral Researcher at University of California, Berkeley - Unofficially work at Berkeley starting July. official starting in Sept. Theoretical physics, **AI**...

Message



Shunyu Yao • 3rd+
Research Scientist at OpenAI
San Francisco, California, United States

Past: Research Intern at MIT-IBM Watson **AI** Lab

Message

Bloomberg.com
https://www.bloomberg.com › news › articles › tencent-a...

Tencent Appoints Former OpenAI Researcher Its Chief AI ...

Dec 17, 2025 — Tencent Holdings Ltd. has appointed **Yao Shunyu** as its chief AI scientist to head its artificial intelligence efforts. · **Yao** will report directly ... Read more

0. A little bit of History: After 2023

After he proposed ReAct, many companies and orgs noticed ReAct's potential. They either started their business or launched open-source projects.

There are too many and I can't name them all here……..

LangChain

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LangChain raises $125M to build the platform for agent engineering

We raised $125M at a $1.25B valuation to build the platform for agent engineering.

3 MIN READ OCT 20, 2025

crewai

CrewAI Launches Multi-Agentic Platform to Deliver on the Promise of Generative AI for Enterprise

CrewAI announces Series A funding, new enterprise offering, and 10 million+ agents automating complex workflows using any LLM and cloud platform

October 22, 2024 08:00 ET | Source: CrewAI

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*Any comments, questions, ideas?

I do have a question for you:

Are you convinced to learn AI Agents?

If you don’t want to learn AI Agent. Why?
Afraid of AI? Sounds like bubble?
Found it useless?

If you want to learn AI Agent. Why?
Found it promising? Found it helpful?
Love money?

Your answer and feedback might impact how we taught AI Agent in the future.

Let me know what you think!

1. Why LangChain?

1. Why LangChain?

(You want to learn others? Let us know! Since they all implemented the ideas of ReAct. Different frameworks have a lot in common. I think I would go to some websites of the following at this moment.)

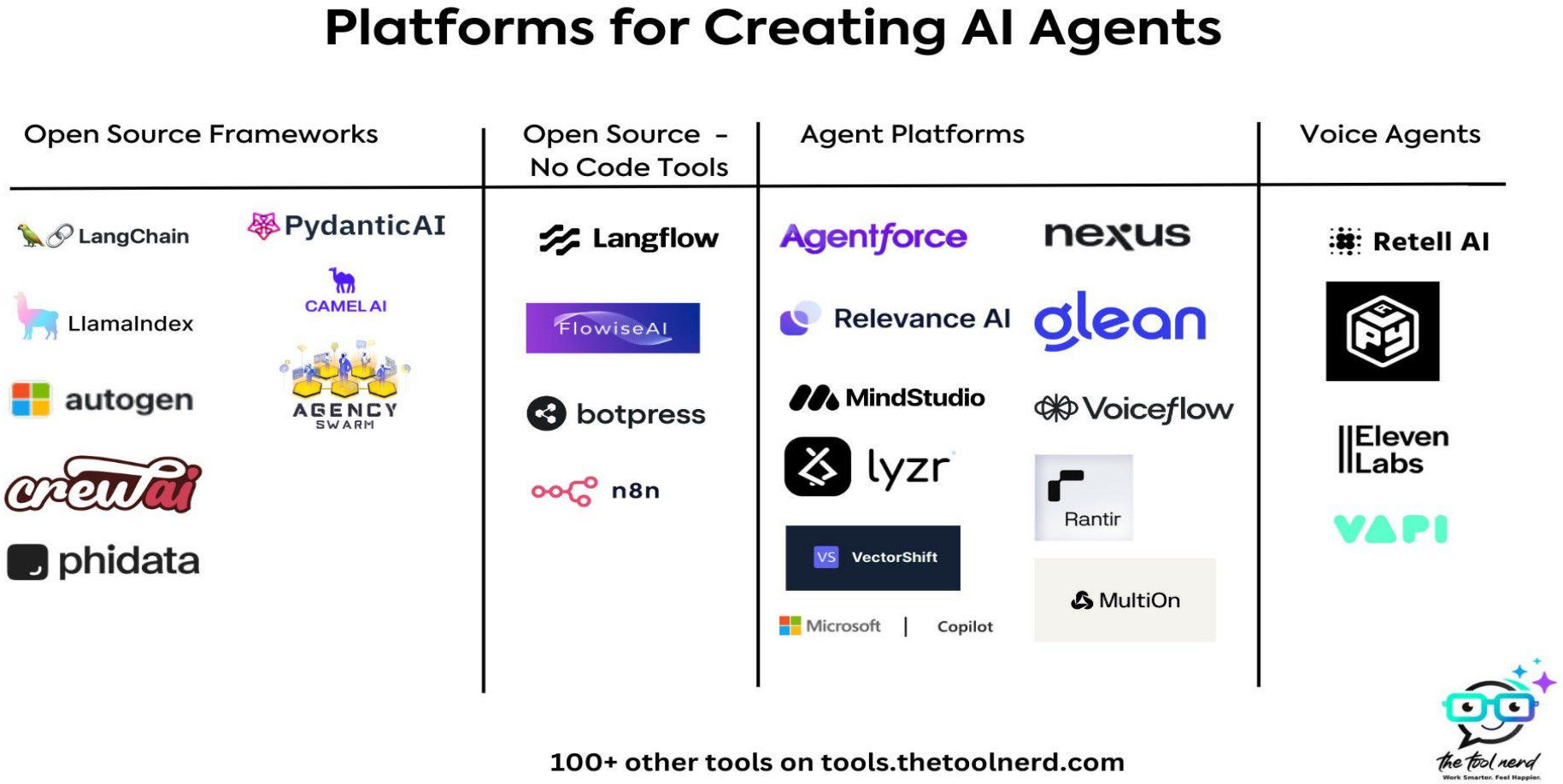


Figure 4: A classification that looks good to me

Figure 4 Source: https://www.thetoolnerd.com/p/best-tools-to-build-ai-agents

Quick Fun Fact:

LangChain

Why LangChain logo is a combination of parrot and chain?

dmyoung9 on Aug 3, 2023

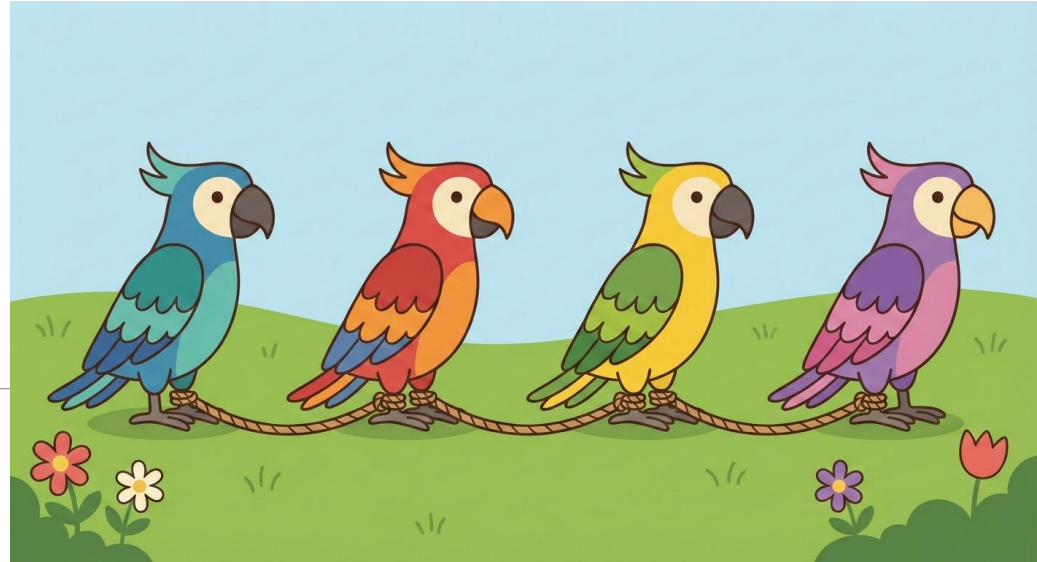
AI Large Language Models have often been compared to or called "stochastic parrots", which is a fancy way of saying they can generate text that mimics human output, but that they don't really have any "understanding" of the generated content.

This library serves to "chain" these "parrots" together in order to get more meaningful and useful outputs.

43 1 2 7

Prompt used to generate image on Nano Banana Pro:

The logo of LangChain consists of a parrot and a chain. The meaning of this logo is that langchain is developed to chain these parrots, the name used in critizing LLM, together to see if it could generate something meaningful. Draw me an image in which parrots are chained together in flat minimal preschool cartoon style. There should be 4 parrots chained by a rope on feet. Different parrots should have different colors. These parrots should be standing in the grassland. a single parrot should have different colors and these colors should coordinate harmoniously. I need to see one side of parrot, not front face. Do not include langchain logo. No sun please.



2. PreReq

2. PreReq

Technical PreReq:

1. uv
2. Python 3.11 or higher
3. LLM API Key (Make sure the OpenAI API key you got works. Let me know if you found any issues)
4. An IDE (it's Integrated Development Environment, not Innovation-Driven Entrepreneurship!!!! I don't care if it's AI-empowered or not. Btw, I am not a fan of colab, unfortunately since I found it hard to use for some reasons. Also, I am quite familiar with 21st century IDE like VS Code and PyCharm, which means if you are using some "old school" IDE like Vim, I can do nothing for you).
5. LangChain v1.2 or higher (TBD. Cause I am not sure if you would need to use v0.3 but virtual env could do this.)
6. Preferably, Git Bash could help me in the future cause I have zero proficiency in MacOS, limited proficiency in Windows command lines and decent proficiency in Linux. Having Git Bash installed could make life easier when dealing with different platforms. Of course, Cygwin is also okay for me.
7. I would use Linux and Win10 in the following presentations.
8. Ability to use OpenAI API key using openai, a Python package maintained by OpenAI Inc.

Non-Technical PreReq:

1……
2……
3…..

(Not much space for it but it doesn't mean it's not important!!!!)

https://git-scm.com/install/

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git --local-branching-on-the-cheap

Cygwin
Get that Linux feeling - on Windows

Thank you for your paged attention!

arXiv > cs > arXiv:2309.06180

Computer Science > Machine Learning

[Submitted on 12 Sep 2023]

Efficient Memory Management for Large Language Model Serving with PagedAttention

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, Ion Stoica